

Classification of DeFi Services using Graph-Based Neural Networks

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Interdisciplinary Project in Data Science

Domain-specific Lecture: TE250 From Idea to Enterprise (completed)

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TL;DR

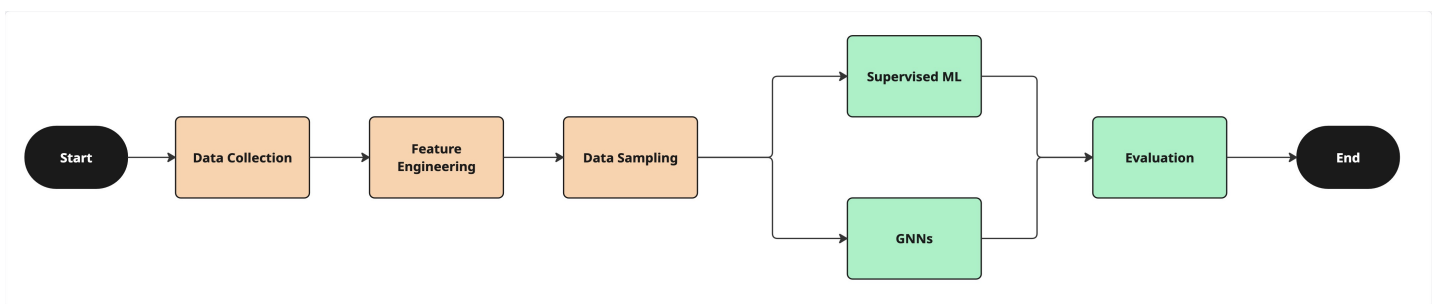
GNNs outperform traditional models for classifying DeFi services by leveraging blockchain graph structure. They scale well and generalize better despite limited, imbalanced data.

Research Questions (RQs)

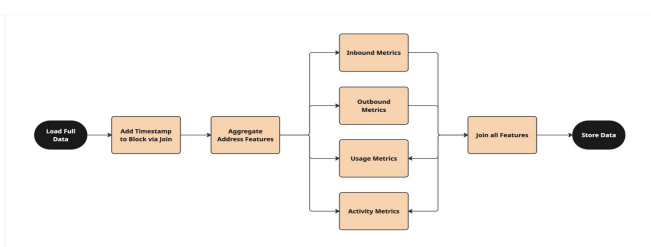
RQ1: To what extent can GNNs be effectively used for entity classification in blockchain networks, and how do they compare to SOTA traditional supervised machine learning methods in terms of accuracy, recall, precision and F1-score?

RQ2: How well do GNNs scale with increasing amounts of input data when, and what are the computational trade-offs compared to traditional ML methods?

Overall Pipeline



Data

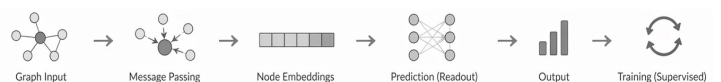


- ~47M Ethereum transactions, ~10M addresses
- Aggregated raw traces into address-level features (inbound, outbound, usage, activity)
- Added custom features like distinct smart contract methods and PageRank
- Limitations:** very few labelled addresses (~1.4k) and strong class imbalance
- Mitigated** via sampling, adding some “unknown” labels, and focusing on top-10 classes

Supervised ML

- Baseline models: Random Forest, Decision Tree, MLP, Logistic Regression
- Input: Engineered address-level features
- Evaluation: 5-fold cross-validation with standard metrics

Graph-Neural-Networks (GNNs)



- Model: GraphSAGE on transaction graph (nodes = addresses, edges = transactions)
- Learning: Message passing to aggregate neighbor information
- Training: Mini-batch neighbor sampling for scalability
- Evaluation: Same split + metrics as baseline for fair comparison

Evaluation & Results

Model	Top-K	Accuracy	Precision	Recall	F1
RandomForest	10	0.7014	0.5791	0.5638	0.5577
GNN	10	0.8326	0.6167	0.5265	0.5586
RandomForest	42	0.4563	0.2999	0.2849	0.2775
GNN	42	0.7624	0.3219	0.2359	0.2527

RQ1:
GNN 🏆 outperforms Random Forest (best) by 13 and 31 percentage points, respectively.

RQ2:

- GNN **scales efficiently** through Neighbor sampling, which keeps memory (~1.5GB) and training time low.
- Better at scale:** Slower on small data, but more efficient for large datasets.

Take-Home Message

- GNNs outperform traditional ML for DeFi classification by leveraging graph structure.
- Strong scalability & generalization** make them suitable for large blockchain datasets.
- Data limitations** (few, imbalanced labels) remain the main performance bottleneck.